

1/27/2025	9 AM - 10 AM	Introduction to NoSQL Databases (e.g., MongoDB)
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<https://www.microsoft.com/en-in/sql-server/sql-server-downloads>

<https://www.mongodb.com/try/download/community-kubernetes-operator>

<https://www.mongodb.com/try/download/shell>

Path:

C:\Program Files\MongoDB\Server\8.0\data\

C:\Program Files\MongoDB\Server\8.0\log\

🌱 MongoDB is an open-source, document-oriented NoSQL database management system.

What is a Document Database?

A document database (also known as a document-oriented database or a document store) is a database that stores information in documents.

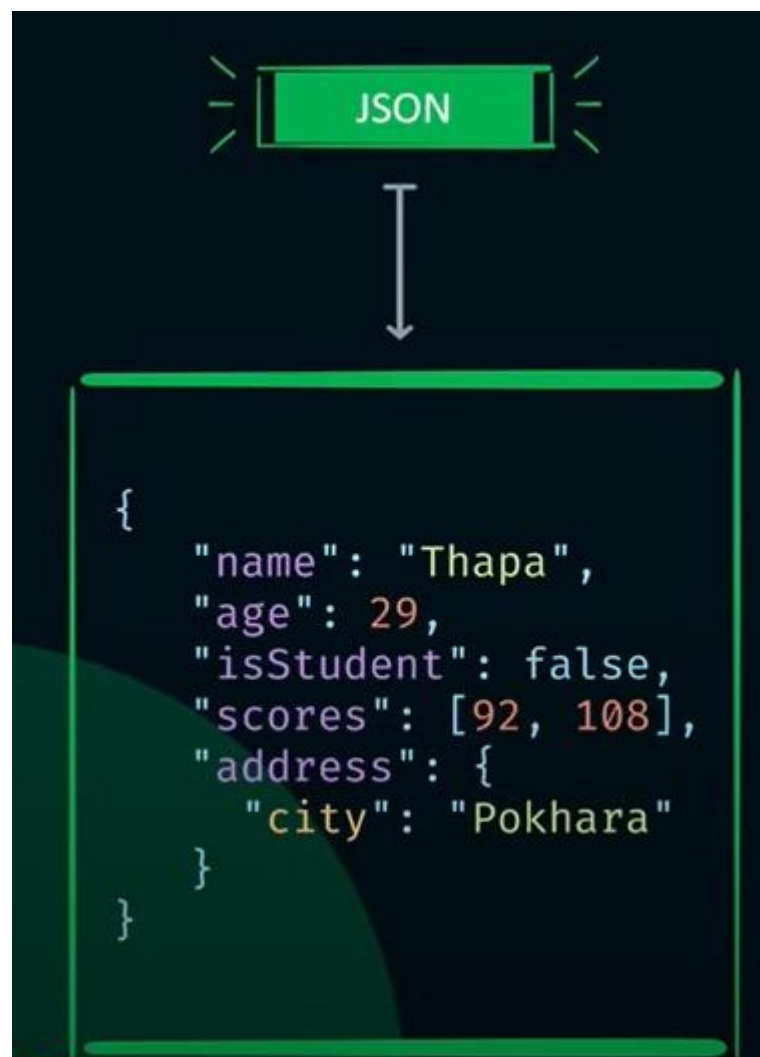
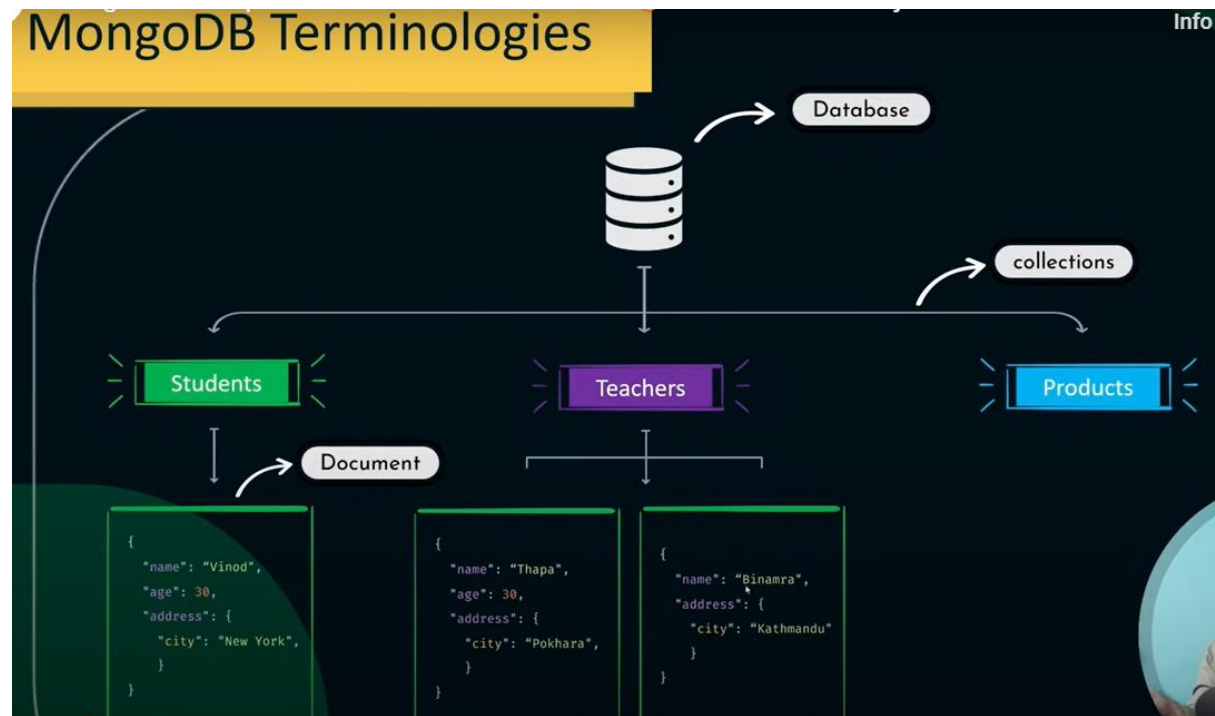
```
{
  "_id": "5cf0029caff5056591b0ce7d",
  "firstname": "Jane",
  "lastname": "Wu",
  "address": {
    "street": "1 Circle Rd",
    "city": "Los Angeles",
    "state": "CA",
    "zip": "90404"
  }
  "hobbies": ["surfing", "coding"]
}
```

SQL	MongoDB(NOSQL)
- SQL databases are relational databases. - They use structured tables to store data in rows and columns. - Suitable for applications with well-defined schemas and fixed data structures. - E-commerce Platform, HR Management etc - Examples: MySQL, PostgreSQL, Oracle.	- NoSQL databases are non-relational databases. - They provide flexibility in data storage, allowing varied data types and structures. - Ideal for applications with dynamic or evolving data models. - CMS, Social Media Platforms, Gaming - Examples: MongoDB, Cassandra,

NOSQL (MONGODB)

Collection: Students
2- documents

```
[
  {
    "_id": "1",
    "first_name": "Vinod",
    "last_name": "Thapa",
    "age": 16,
    "grade": 11,
    "subjects": [
      { "subject_name": "Mathemati",
        { "subject_name": "Computer"
      }
    ]
  },
  {
    "_id": "2",
    "first_name": "Thapa",
    "last_name": "Technical",
    "age": 17,
    "grade": 12,
    "extra": 'sport captain',
    "subjects": [
      { "subject_name": "Mathemati",
        { "subject_name": "Computer"
      }
    ]
  }
]
```

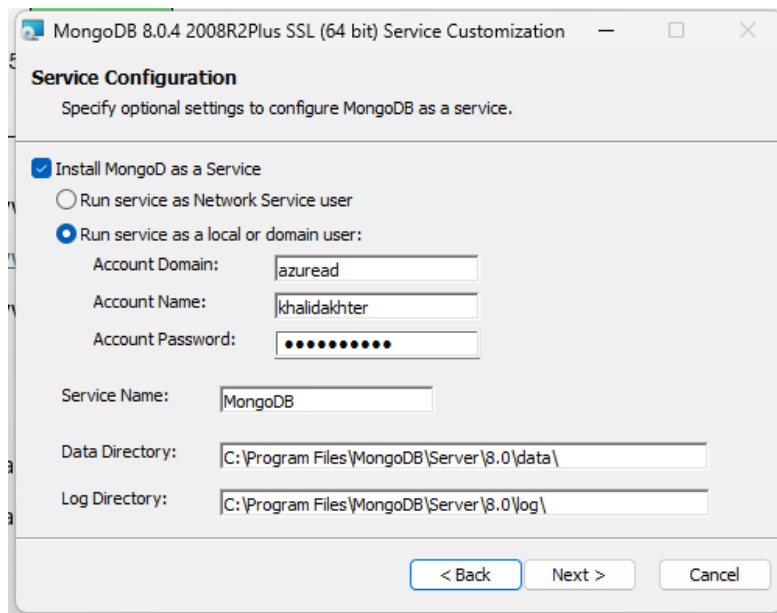


Open cmd and write

Whoami

For account domain and account name

Installation of MongoDB



use master

exec sp_dropdistributor @no_checks=1, @ignore_distributor=1

mongod --version

start the server

>mongosh to check the shell

show dbs

use <database>

db.dropDatabase()

show collection;

db.createCollection<collection-name>) => equivalent to Table

db.<collection-name>.drop

```
test> show dbs
```

```
admin  40.00 KiB
```

```
config 60.00 KiB
```

```
local  40.00 KiB
```

```
test> show databases
```

```
admin  40.00 KiB
```

```
config 60.00 KiB
```

```
local  40.00 KiB
```

```
test> use students
```

```
switched to db students
```

```
students> show dbs
```

```
admin  40.00 KiB
```

```
config 60.00 KiB
```

```
local  40.00 KiB
```

```
students> use admin
```

```
switched to db admin
```

```
admin> show dbs
```

```
admin  40.00 KiB
```

```
config 60.00 KiB
```

```
local  40.00 KiB
```

```
admin> use students
```

```
switched to db students
```

```
students> show collections
```

```
students> db.createCollection('data')
```

```
{ ok: 1 }
```

```
students> show dbs
```

```
admin  40.00 KiB
```

```
config 60.00 KiB
```

```
local  40.00 KiB
```

```
students 8.00 KiB
```

```
students>
```

```
students> show collections
```

data

students>

students> show collections

data

students> db.data.drop()

true

students> show collections

students>

students> db.dropDatabase()

{ ok: 1, dropped: 'students' }

students> show dbs

admin 40.00 KiB

config 92.00 KiB

local 40.00 KiB

students>

students> db.data.insertOne({'name':'Khalid',age:29})

{

acknowledged: true,

insertedId: ObjectId('67853eeebaea1c5776cb0ce2')

}

students>

students>

db.data.insertMany([{'name':'Akhter',age:29},{ 'name':'rajesh',age:30},{ 'name':'Rohit',age:68}])

{

acknowledged: true,

insertedIds: {

'0': ObjectId('67853fd4baea1c5776cb0ce3'),

'1': ObjectId('67853fd4baea1c5776cb0ce4'),

'2': ObjectId('67853fd4baea1c5776cb0ce5')

}

}

students>

students> db.data.find()

[

```
{
  _id: ObjectId('67853eeebaea1c5776cb0ce2'),
  name: 'Khalid',
  age: 29
},
{
  _id: ObjectId('67853fd4baea1c5776cb0ce3'),
  name: 'Akhter',
  age: 29
},
{
  _id: ObjectId('67853fd4baea1c5776cb0ce4'),
  name: 'rajesh',
  age: 30
},
{ _id: ObjectId('67853fd4baea1c5776cb0ce5'), name: 'Rohit', age: 68 },
{
  _id: ObjectId('67854007baea1c5776cb0ce6'),
  name: 'Akhter',
  age: 29
},
{
  _id: ObjectId('67854007baea1c5776cb0ce7'),
  name: 'rajesh',
  age: 30
},
{ _id: ObjectId('67854007baea1c5776cb0ce8'), name: 'Rohit', age: 68 }
]
students>
```

```
students> db.data.insertOne({'name':'Khalid',age:29,'course name':'Computer Science'})
```

```
{
  acknowledged: true,
  insertedId: ObjectId('6785412bbaea1c5776cb0ce9')
}
students>
```

```
students> db.data.find().limit(2)
```

```
students> db.data.find({'name':'Khalid'})
```

```
[
{
  _id: ObjectId('67853eeeebaea1c5776cb0ce2'),
  name: 'Khalid',
  age: 29
},
{
  _id: ObjectId('6785412bbaea1c5776cb0ce9'),
  name: 'Khalid',
  age: 29,
  'course name': 'Computer Science'
}
]
students>
```

```
students> db.data.findOne({'name':'Khalid'})
{ _id: ObjectId('67853eeeebaea1c5776cb0ce2'), name: 'Khalid', age: 29 }
students>
```

```
students> db.data.find({'name': 'Khalid'}).count()
2
students>
```

```
students>
db.data.updateOne({_id:ObjectId('6785412bbaea1c5776cb0ce9')},{ $unset:{'name':1}})
--Remove field
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
students> db.data.deleteOne({name:'rajesh'})
{ acknowledged: true, deletedCount: 1 }
```

```
students> db.data.find().count()
7
students> db.data.deleteOne({name:'rajesh'})
{ acknowledged: true, deletedCount: 1 }
```



```
students> db.data.find().count()
```

```
6
```

```
students>
```

```
students> db.data.updateOne({name:'Rohit'}, {$set: {name:'Rohit Kumar'}})
```

```
{
```

```
  acknowledged: true,
```

```
  insertedId: null,
```

```
  matchedCount: 1,
```

```
  modifiedCount: 1,
```

```
  upsertedCount: 0
```

```
}
```

```
students> db.data.updateOne({name:'Rohit Kumar'}, {$set: {name:'Kumar3'}})
```

	3 PM - 4 PM	Installation of Visual Studio Community Version Create Console Program in C#
	4 PM - 5 PM	Classess in C# Static Class Sealed Class Abstract Class Normal Class Interface

	5 PM - 6 PM	Experince Talk - Intern 2024
	6 PM - 7 PM	Practice Session